

UST'-NYUKZHA, Russian Federation

WMO: 303850

Lat: 56.560N

Lon: 121.591E

Elev: 426

StdP: 96.32

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.7	-39.7	-43.5	0.1	-41.5	-41.5	0.1	-39.5	4.7	-25.4	3.5	-25.7	0.1	340	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	29.3	19.0	27.3	18.4	25.2	17.5	20.5	26.7	19.4	25.3	18.4	23.6	0.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.0	22.9	17.4	13.1	21.9	16.4	12.3	21.0	61.3	26.6	57.6	25.5	54.1	23.6	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.5	2.9	2.3	-43.9	32.7	2.3	1.6	-45.5	33.8	-46.9	34.8	-48.2	35.6	-49.9	36.8		
(5)			-43.8	22.8	2.2	1.3	-45.4	23.8	-46.7	24.6	-47.9	25.3	-49.5	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.7	-30.1	-23.7	-13.2	-1.3	7.6	15.9	18.4	15.7	7.3	-3.9	-19.6	-30.0	
	DBStd	18.10	5.26	5.21	6.20	4.69	4.03	3.80	3.21	3.70	4.14	5.58	7.23	5.57	
	HDD10.0	6014	1244	942	718	341	98	4	1	4	102	432	888	1241	
	HDD18.3	8462	1502	1176	976	591	334	91	38	96	332	690	1138	1499	
	CDD10.0	666	0	0	0	0	22	180	262	181	21	0	0	0	
	CDD18.3	72	0	0	0	0	0	16	41	15	0	0	0	0	
	CDH23.3	894	0	0	0	0	14	271	426	181	3	0	0	0	
	CDH26.7	239	0	0	0	0	3	76	123	37	0	0	0	0	
Wind	WSAvg	0.5	0.4	0.3	0.5	0.7	0.7	0.5	0.5	0.4	0.4	0.5	0.4	0.4	
Precipitation	PrecAvg	505	8	5	5	19	44	94	117	101	70	22	11	8	
	PrecMax	694	17	12	19	40	89	221	276	253	171	78	35	24	
	PrecMin	308	2	0	0	2	19	23	24	20	21	9	4	2	
	PrecStd	103	4	3	4	11	18	48	60	45	38	14	7	4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-13.8	-6.4	5.2	15.8	25.0	31.9	32.2	29.9	23.1	11.0	-0.9	-11.3	
		MCWB	-14.7	-8.3	1.2	7.8	13.7	19.9	20.6	19.2	15.0	5.7	-2.7	-11.9	
	2%	DB	-17.5	-10.1	1.9	11.8	21.4	28.8	29.7	27.4	19.6	7.8	-4.6	-15.6	
		MCWB	-18.1	-11.4	-1.5	5.3	12.0	17.9	19.5	18.5	12.8	3.5	-5.9	-16.3	
	5%	DB	-20.2	-13.4	-0.6	9.0	18.7	26.4	27.8	25.3	16.9	5.6	-6.9	-19.9	
		MCWB	-20.6	-14.3	-3.3	3.4	10.6	17.0	19.1	17.8	11.7	2.1	-8.0	-20.2	
	10%	DB	-22.7	-15.7	-2.8	6.5	15.9	24.1	25.6	22.9	14.5	3.5	-9.3	-22.5	
		MCWB	-23.0	-16.5	-5.1	1.7	8.9	16.1	18.3	16.8	10.2	0.8	-10.3	-22.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-14.7	-8.3	1.1	8.8	14.6	21.4	22.3	21.0	16.0	5.9	-2.3	-12.0
			MCDB	-13.8	-6.7	4.8	14.4	23.2	29.6	29.8	26.7	20.8	10.3	-0.9	-11.4
2%		WB	-18.0	-11.5	-1.3	5.5	12.6	19.5	20.9	19.7	14.0	4.0	-5.8	-16.3	
		MCDB	-17.5	-10.3	1.6	11.2	20.2	26.8	26.9	25.2	18.0	7.1	-4.6	-15.6	
5%		WB	-20.6	-14.2	-3.3	3.7	11.1	18.1	19.8	18.6	12.2	2.4	-8.0	-20.2	
		MCDB	-20.2	-13.4	-0.7	8.4	17.3	24.4	25.5	23.6	16.2	5.2	-7.0	-19.9	
10%		WB	-23.0	-16.5	-5.1	2.0	9.5	16.7	18.9	17.5	10.6	0.9	-10.3	-22.7	
		MCDB	-22.7	-15.7	-2.8	6.0	14.9	22.4	24.4	21.7	13.7	3.3	-9.3	-22.5	
Mean Daily Temperature Range		5% DB	MDBR	7.5	11.2	13.7	11.8	12.1	13.3	12.0	11.7	10.6	8.9	8.3	6.7
			MCDBR	8.4	12.4	14.2	15.1	18.0	18.1	16.4	16.1	15.3	11.0	9.4	8.2
	MCWBR		8.0	11.3	11.1	9.4	9.7	8.3	7.3	7.7	8.6	7.2	7.9	7.8	
	5% WB	MCDBR	8.4	12.3	13.9	14.0	16.2	15.6	13.6	13.7	13.6	10.4	9.2	8.2	
		MCWBR	7.9	11.2	11.0	9.0	9.3	8.1	6.8	7.3	8.1	7.0	7.8	7.8	
Clear-Sky Solar Irradiance	taub	0.253	0.285	0.317	0.410	0.496	0.469	0.482	0.420	0.349	0.310	0.265	0.244		
	taud	2.036	2.033	2.016	1.885	1.874	2.102	2.088	2.262	2.404	2.282	2.074	1.980		
	Ebn at Noon	617	750	824	789	742	777	755	784	801	729	590	505		
	Edh at Noon	61	94	125	168	182	147	147	115	86	73	58	49		
All-Sky Solar Radiation	RadAvg	0.79	1.77	3.42	5.11	5.16	5.41	5.01	4.14	2.84	1.76	0.97	0.54		
	RadStd	0.05	0.09	0.19	0.27	0.49	0.43	0.43	0.35	0.27	0.12	0.05	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	+1.67	+1.68	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page